

Machine Learning-Based Prediction of HER2 Status Using DCE-MRI Radiomic Features

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I. Abstract

HER2+ breast tumors are aggressive, have a higher recurrence rate, and increased mortality, yet determining HER2 status relies on a multi-step biopsy process. This report investigates whether quantitative DCE-MRI radiomic features can predict HER2 status in a cohort of 922 patients with invasive breast cancer. Three machine learning classifiers, logistic regression (LR), random forest (RF), and support vector machine (SVM), were trained on each of nine radiomic feature groups, with AUC as the primary performance metric and 95% confidence intervals estimated using 1,000 bootstrap iterations. Tumor enhancement achieved the highest performance for LR and SVM (AUC = 0.621 and 0.641), while tumor size and morphology performed best for RF (AUC = 0.618); no classifier exceeded an AUC of 0.65. While no AUC values reached a clinically significant threshold, performance above chance suggests stronger classification is achievable with proper model refinement.

II. Introduction

Breast cancer remains one of the most frequently diagnosed cancer types amongst women globally. In 2022, 2.3 million new cases were found and 670,000 deaths occurred.^{1,2} By 2050, it is expected that we will see a 38% increase in new cases and a 68% increase in deaths.² While survival rates have improved due to advances in screening and treatment, outcomes remain highly dependent on early detection and accurate tumor characterization.³ These statistics highlight the increasing burden of the disease and the continued need to invest in research aimed at improving diagnosis, treatment, and patient outcomes.

Molecular subtypes provide insight into the aggressiveness and treatment plan of a tumor. These subtypes are defined by the expression of the estrogen receptor (ER), progesterone

receptor (PR), and human epidermal growth factor receptor 2 (HER2), resulting in four classifications: Luminal A, Luminal B, HER2-enriched, and triple negative.^{4,5} Among these subtypes, HER2+ tumors are of particular interest due to their aggressiveness, higher recurrence rate, and increased mortality. HER2 is a receptor tyrosine kinase whose overexpression leads to the dysregulation of key cell signaling pathways, driving uncontrolled tumor cell proliferation and growth.⁶ Occurring in approximately 20% of invasive breast cancers, overexpression of HER2 has become an important biomarker for guiding treatment decisions.⁷⁻⁹ Despite the rapid growth and progression of these tumors, the success of HER2-targeting therapies including Trastuzumab, tyrosine kinase inhibitors (TKIs) and antibody-drug conjugates (ADCs) provides a basis for treatability.^{10,11}

Accurate determination of HER2 status is necessary for guiding treatment, yet obtaining this information relies on a multi-step diagnostic process. HER2 status is determined through a biopsy or removal of a tumor and either an immunohistochemistry (IHC) test or fluorescence in situ hybridization (FISH) test is performed to determine the level of HER2 protein in the cancerous cells.⁹ While HER2 testing has become standard practice following tumor identification, the time required to complete a biopsy and return results introduces delays that are particularly concerning given the rapid progression of HER2+ tumors.^{11,12} As a result, identifying such molecular characteristics earlier in the diagnostic process may provide important clinical value. This gives us reason to explore whether medical imaging techniques can help predict HER2 status before biopsy results become available.

Several existing studies have investigated the application of machine learning models to medical imaging for predicting clinical outcomes in breast cancer. Extraction of quantitative features from imaging modalities such as magnetic resonance imaging (MRI), positron emission

tomography (PET), and computed tomography (CT) have advanced our ability to more precisely characterize tumors.¹³ While existing imaging modalities such as MRI, PET, and CT provide valuable structural information, dynamic contrast-enhanced MRI (DCE-MRI) offers a combination of high spatial resolution, high temporal resolution, further functional insights, and contrast enhancement dynamics, making it especially well-suited for breast tumors.^{14,15} Previous studies have demonstrated promising results using DCE-MRI radiomics for molecular subtype prediction,¹⁶ however, existing work has been limited to smaller sample sizes, limited exploration of DCE-MRI applications, and a lack of validity across various models.

This report seeks to address these gaps by investigating whether quantitative DCE-MRI radiomic features can predict HER2 status in a cohort of 922 patients. Building on prior radiogenomic analyses of this dataset,¹⁶ three machine learning classifiers were applied across nine radiomic feature groups to identify which categories of imaging features may be most predictive of HER2 status.

III. Materials and Methods

Dataset

Clinical and imaging data was accessed from a publicly available dataset of 922 female patients with invasive breast cancer and pre-operative DCE-MRI via The Cancer Imaging Archive.¹⁶ Data was collected and compiled from the Duke University Medical Center between January 2000 and March 2014. Full details of patient inclusion and exclusion criteria, imaging acquisition parameters, and pathological assessment are described in Saha et al. 2018.¹⁷ Two data sources were utilized in this study: a clinical dataset containing 98 variables, and an imaging features dataset containing 529 extracted DCE-MRI radiomic features.

Clinical Data

A ‘Clinical and Other Features’ dataset contained 12 categories: demographics, tumor characteristics, MRI findings, surgery, radiation therapy, tumor response, recurrence, follow-up, mammography characteristics, ultrasound features, therapy data, and MRI technical information. HER2 status, the primary outcome variable of this report, was identified within tumor characteristics and was obtained from a pathology biopsy report.

Imaging Data

529 imaging features were extracted from the tumor and automatically segmented fibroglandular tissue (FGT) and organized into 10 feature groups based on their source of extraction and the imaging characteristics they capture, as described in Saha et al. 2018. A supporting figure from the paper is included in the Appendix to better depict the feature group organization (Figure A).

The ‘Combining tumor and FGT enhancement’ group was excluded due to inconsistent naming conventions in the given supplemental documentation and the imaging dataset. Therefore, the assigned features could not be identified for this group. The remaining nine groups included 511 features which were retained for analysis. A breakdown of the feature groups and their descriptions is provided in the Appendix (Table A).

Data Preprocessing

Missing Data

HER2 status was complete across all 922 patients with no missing values. Of the 529 imaging features, 40 contained at least one missing value, accounting for only 547 missing entries out of 487,838 total (0.11% of the dataset). The most affected feature was missing in 44 patients (4.77% of the cohort). Given the low overall rate of missing values, median imputation was applied to all affected features. Median imputation was selected due to its robustness to outliers.

Given that missing values represented a small percentage of the dataset, imputation is unlikely to have introduced meaningful bias into the analysis.

Train/Test Split

An 80/20 split was used to maximize the amount of training data available for each model. The training set included 737 patients and the testing set included 185 patients. The dataset exhibited a class imbalance, with HER2+ cases comprising 163 patients (17.7% of the cohort). To account for this, stratified splitting was applied, ensuring that the proportion of HER2+ patients was consistent for both sets, resulting in a distribution of 130 HER2+ patients (17.6%) in the training set and 33 HER2+ patients (17.8%) in the test set.

Machine Learning Models

The primary goal of the project was to determine whether machine learning models could effectively classify whether a tumor was HER2+ or HER2– based on the features in the feature group. Three machine learning models were evaluated individually and compared with one another for validity of the results: logistic regression (LR), random forest (RF), and support vector machine (SVM). All three models were trained independently on each of the nine radiomic feature groups. Prior to training, features were standardized using standard scaler to ensure that features with larger numerical ranges did not dominate the model. Additionally, to account for the class imbalance in the dataset, class weights were set to balanced across all models. Each model was trained on the training set and evaluated on the test set using AUC as the primary performance metric.

Logistic regression was applied as a baseline model due to its interpretability and simplicity. A maximum of $n = 1,000$ iterations was set to allow the model to fully converge. Random forest was applied to capture non-linear relationships between features otherwise

detected by a linear model. The model was configured with $n = 200$ decision trees to ensure a stable and reliable prediction. Support vector machine was applied using a radial basis function (RBF) kernel to capture the non-linear relationships across features.

Statistical Analysis

Statistical analysis was conducted to validate model performance beyond the single point estimate AUC value. A total of 1,000 bootstrap samples were drawn from the test set with replacement to construct 95% confidence intervals for each feature group and model combination. In each iteration, 185 patients were randomly sampled from the test set with replacement. Therefore, some patients could be selected multiple times while others were not selected at all, producing slightly varying test sets each time. The bounds of the confidence interval were determined at the 2.5 and 97.5 percentiles. The model was trained only once on the training set, and bootstrapping was applied only to the test set evaluation.

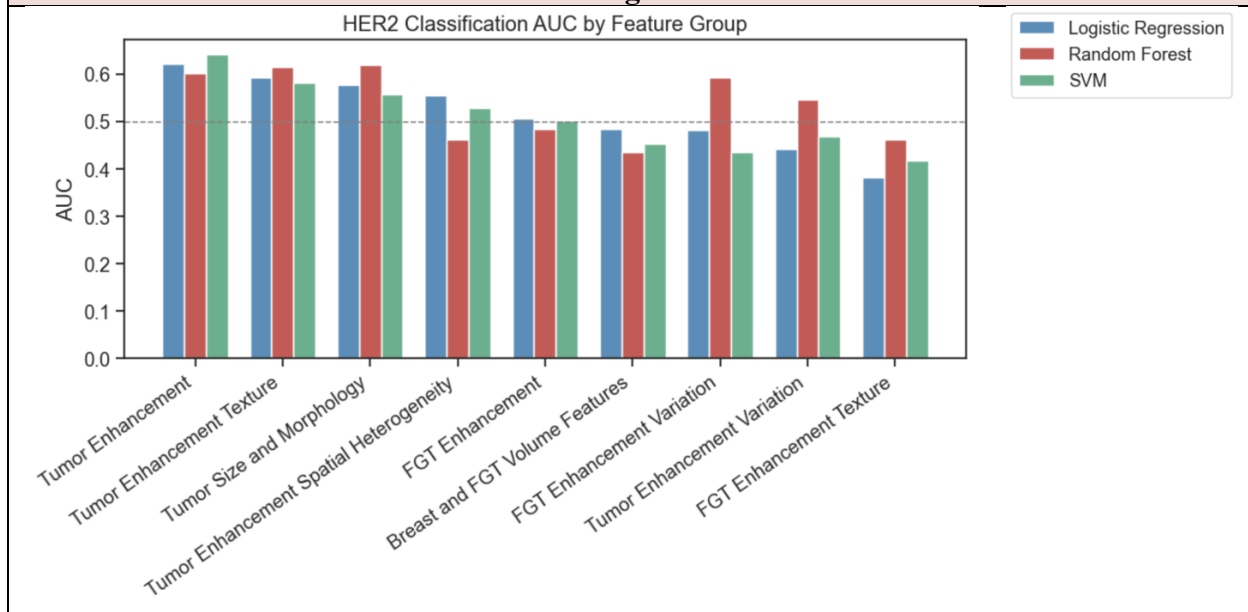
IV. Results

The AUC for HER2 classification was reported across the three machine learning models for the nine radiomic feature groups (Table 1 and Figure 1). Tumor enhancement achieved the highest classification performance for LR and SVM, with AUC values of 0.621 and 0.641 respectively. Tumor size and morphology performed best for RF, with a maximum AUC value of 0.618. In comparison, several FGT-derived feature groups performed near or below chance level (AUC = 0.5). The worst performing model was the LR model on FGT enhancement texture, with a minimum AUC value of 0.381.

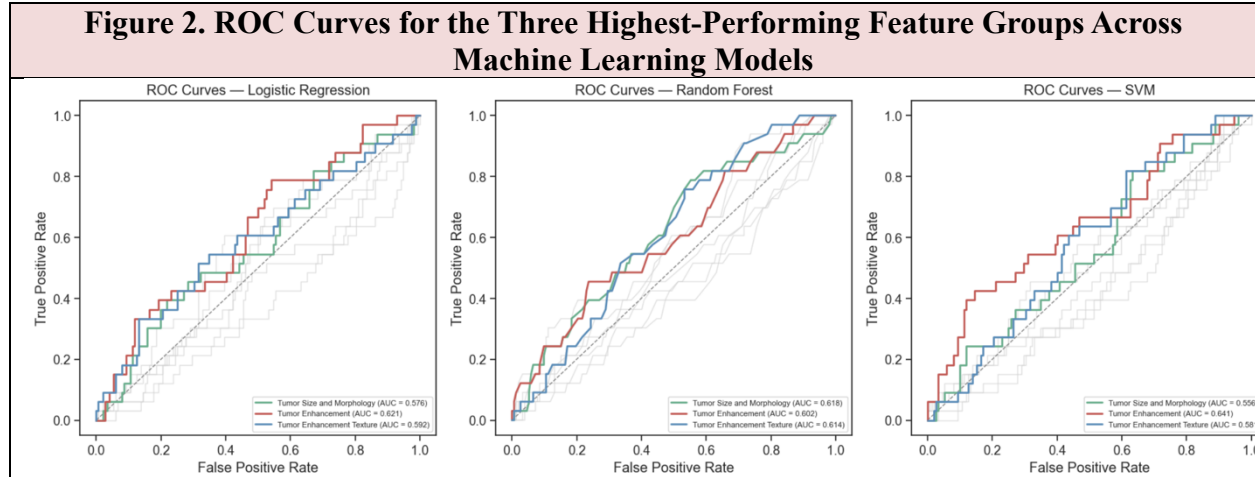
Table 1. AUC for HER2 Status Prediction Across Feature Groups and Machine Learning Models

Feature Group	N Features	LR	RF	SVM
Breast and FGT volume features	5	0.484	0.434	0.453
Tumor size and morphology	10	0.576	0.618	0.556
FGT enhancement	82	0.505	0.483	0.502
Tumor enhancement variation	31	0.442	0.593	0.468
FGT enhancement texture	176	0.381	0.462	0.418
Tumor enhancement texture	135	0.592	0.614	0.581
Tumor enhancement spatial heterogeneity	4	0.555	0.462	0.528
FGT enhancement variation	34	0.481	0.593	0.434
Tumor enhancement	30	0.621	0.546	0.641

LR = Logistic Regression, RF = Random Forest, SVM = Support Vector Machine

Figure 1. Comparison of HER2 Classification AUCs Across Feature Groups and Machine Learning Models

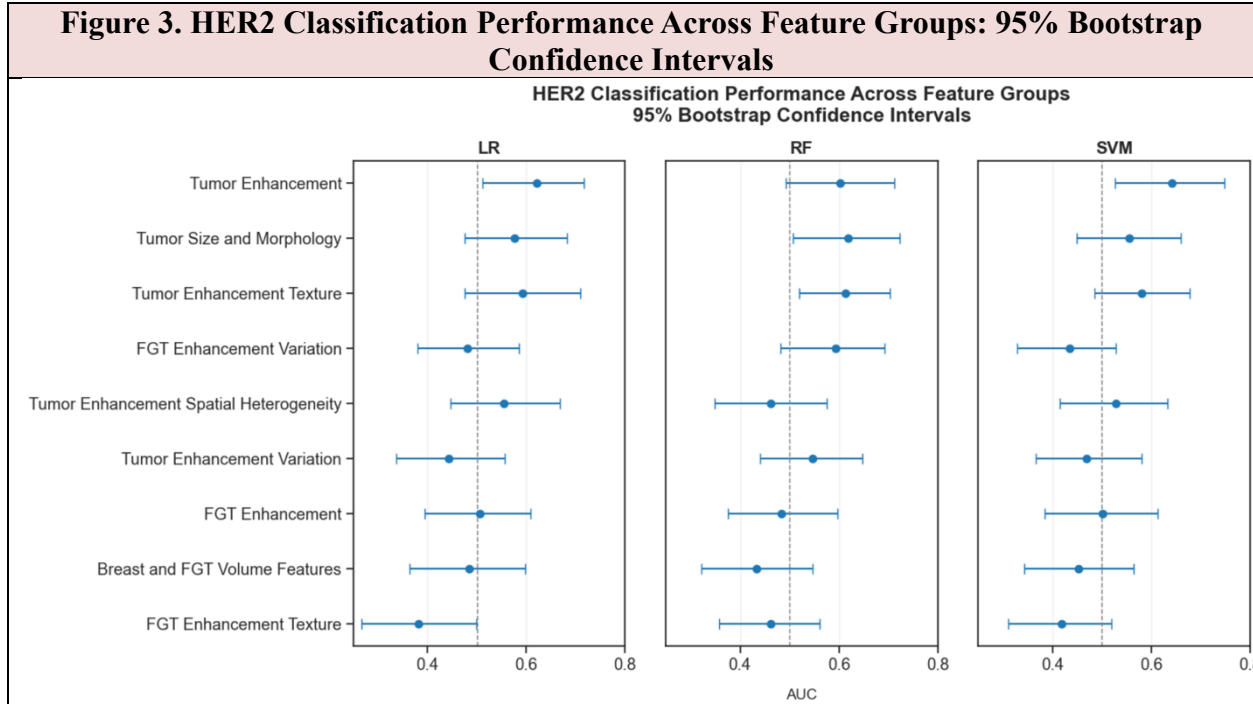
Receiver operating characteristic (ROC) curves for the three highest-performing feature groups are shown in Figure 2. The ROC curves also illustrate that model performance was only modest, as no classifier achieved an AUC greater than 0.65.



To assess the stability of model performance, 95% confidence intervals were estimated using 1,000 bootstrap iterations and are reported in Table 2 and visualized in Figure 3. The highest-performing feature groups generally maintained confidence intervals centered above 0.5. Tumor enhancement produced confidence intervals of 0.511–0.717, 0.492–0.712, and 0.527–0.747 for LR, RF, and SVM, respectively. Tumor size and morphology and tumor enhancement texture also exhibited confidence intervals above 0.5 across models.

Table 2. 95% Bootstrap Confidence Intervals for HER2 Classification Performance Across Feature Groups and Machine Learning Models			
Feature Group	LR	RF	SVM
Breast and FGT volume features	(0.363–0.598)	(0.321–0.546)	(0.342–0.564)
Tumor size and morphology	(0.475–0.682)	(0.507–0.723)	(0.449–0.659)
FGT enhancement	(0.395–0.609)	(0.375–0.598)	(0.384–0.612)
Tumor enhancement variation	(0.336–0.556)	(0.440–0.648)	(0.366–0.580)
FGT enhancement texture	(0.267–0.499)	(0.358–0.562)	(0.311–0.519)
Tumor enhancement texture	(0.476–0.710)	(0.520–0.703)	(0.485–0.678)
Tumor enhancement spatial heterogeneity	(0.446–0.668)	(0.349–0.576)	(0.415–0.633)
FGT enhancement variation	(0.379–0.585)	(0.481–0.693)	(0.329–0.529)
Tumor enhancement	(0.511–0.717)	(0.492–0.712)	(0.527–0.747)

LR = Logistic Regression, RF = Random Forest, SVM = Support Vector Machine



V. Discussion

The feature groups with the highest AUC performance are consistent with what might be expected biologically. Tumor-specific features may be better predictors than FGT features since they directly reflect characteristics of the tumor itself rather than the surrounding tissue. Among the tumor features, the enhancement and size-related groups performed strongest. This aligns with the known biology of HER2+ tumors as their rapid proliferation, increased vascularity, and aggressive growth patterns may be more directly captured through the contrast enhancement dynamics indicated by DCE-MRI.

While the results offered some insight into which feature groups were better at classifying HER2 status, it is important to acknowledge that none of the AUC values reached a threshold typically considered clinically meaningful, which is generally above 0.8. This makes sense given that HER2 status is ultimately determined at the molecular level, and imaging features can only capture it indirectly. However, the feature groups' ability to perform above

chance suggests that with model improvements, classification performance could be strategically improved. This offers a promising direction for future use.

Limitations of this study include the use of pre-collected data, which limited control over data collection procedures and cohort stratification. Although missing data were not extensive, continued efforts to investigate the missing data should be conducted to determine whether specific patients, imaging features, or imaging groups should be examined more closely.. In addition, no hyperparameter tuning was performed, leaving opportunities for further optimization and potential improvements in predictive performance.

Future work can be pursued in both the short and long term. More immediately, hyperparameter tuning, additional machine learning models, alternative train/test splits, and cross-validation could all be explored to improve model performance. Long term studies may include feature-level analysis within the top-performing groups, joint integration of clinical and imaging predictors, and extensions of machine learning to multi-class classifications or continuous variables.

Overall, the results of this study provide a foundation for understanding which categories of DCE-MRI imaging features carry the most predictive signal for HER2 status in breast cancer. Tumor enhancement emerged as the most consistent and statistically significant group across all three classifiers, suggesting that tumor enhancement dynamics may have the most promise for a non-invasive, imaging-based HER2 prediction. This project serves as an initial exploratory baseline for the possibilities of machine learning and DCE-MRI, with many more possible routes of future exploration.

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Appendix

Supplementary code can be accessed via [GitHub](#)

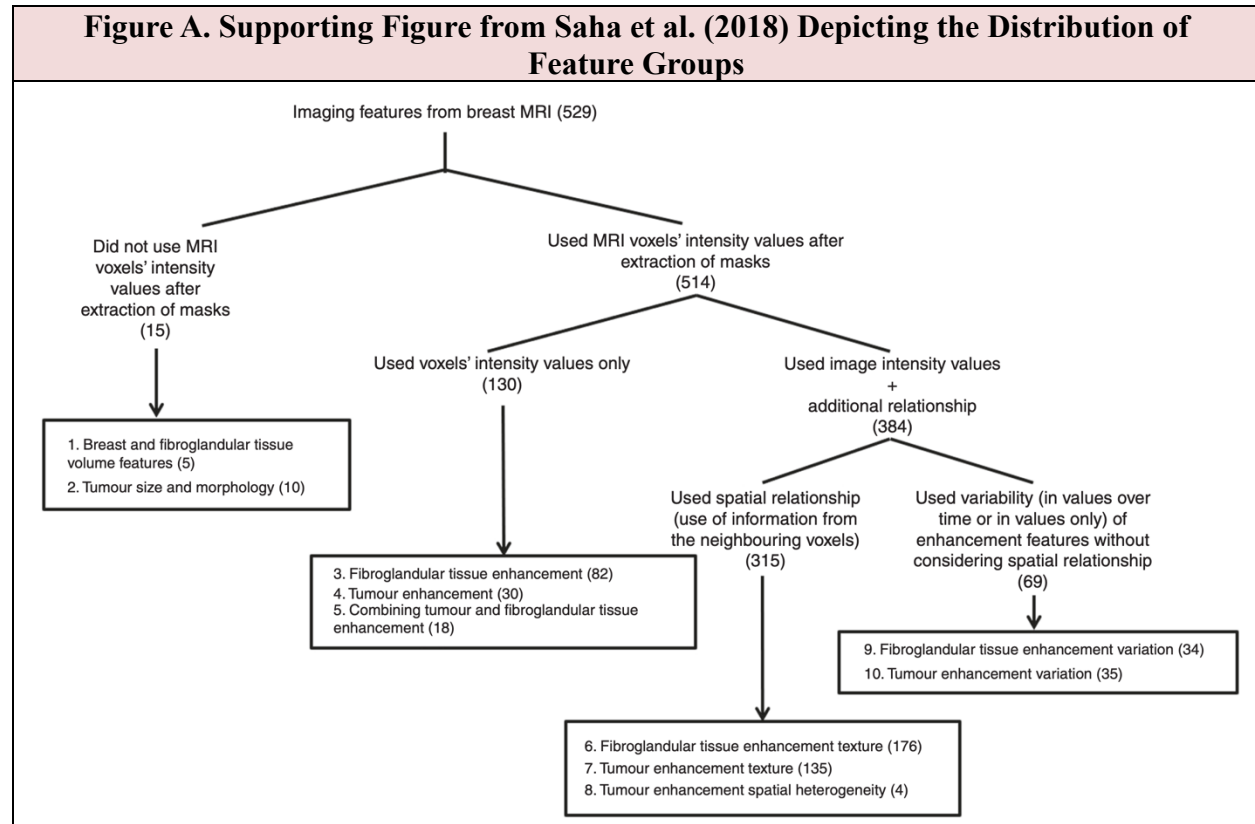


Table A. Breakdown of Image Feature Groups Extracted from DCE-MRI, as Described in Saha et al. (2018)

Feature Group	Description
Breast and FGT volume features	Quantifies the overall volume of the breast and the FGT within it.
Tumor size and morphology	Captures the size, shape, and structural characteristics of the segmented tumor.
FGT enhancement	Describes the degree to which the FGT enhances following contrast administration.
Tumor enhancement variation	Describes the degree to which the tumor enhances following contrast administration.
FGT enhancement texture	Captures the texture, or spatial pattern, of enhancement within the FGT.
Tumor enhancement texture	Captures the texture, or spatial pattern, of enhancement within the tumor.
Tumor enhancement spatial heterogeneity	Captures how unevenly enhancement is distributed across the tumor.
FGT enhancement variation	Captures variation in enhancement kinetics, or the rate of contrast uptake and washout, within FGT.
Tumor enhancement	Captures variation in enhancement kinetics, or the rate of contrast uptake and washout, within tumor.